10 A shop sells

17 different types of sandwich
15 different types of drink
26 different types of snack
14 different types of dessert.

A meal deal consists of either

a sandwich, a drink and a snack
or a sandwich, a drink and a dessert.

(a) Show that there are over 10 000 different ways to choose a meal deal.

(3)

The owner of the shop says,

> "If I halve the number of snacks available, this will halve the number of ways to choose a meal deal."

The owner of the shop is incorrect.

(b) Explain why.

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(1)

(Total for Question 10 is 4 marks)